MA6217 Project Report

Group 7
Liu Bingnan, Vilayanur Adithya Haridas, Xu Siyu, Yang Jiaxin, Zhang Mengliu

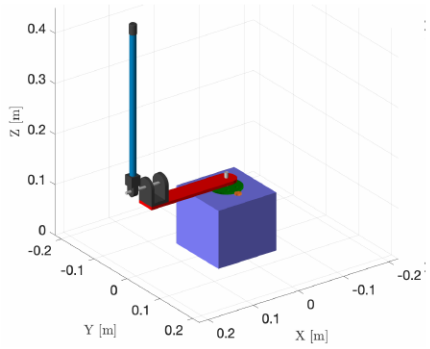


Fig. 1. Rotary inverted pendulum system.

***Abstract*—This report designs a linear-quadratic-Gaussian (LQG) controller that balances a Quanser rotary inverted pendulum and tracks arm-angle references. A four-state linear model is derived and shown to be controllable and observable. A Kalman observer estimates the angular velocities from the two measured angles, an LQR regulator is selected from four weight sets, and a static feedforward gain $N_{\text{bar}} = -3.1623$ gives zero steady-state arm error for the nominal model. Separate observer and regulator designs are justified through the block-triangular closed-loop matrix. In a nonlinear Simulink simulation the arm tracks commands of 20°, -15°, 30° and 0° with final-second error below 10^{-3} degrees and a peak motor voltage of 2.48 V, inside the 12 V limit.**

***Index Terms*—LQG control, Kalman observer, LQR weight selection, feedforward reference tracking, rotary inverted pendulum**

I. Introduction

The rotary inverted pendulum is unstable at its upright equilibrium, and keeping it balanced while repositioning the rotary arm requires feedback control. The arm and pendulum angles are measured directly, while their angular velocities are estimated using a Kalman observer. An LQR controller is first designed to stabilize the pendulum, then combined with a reference feedforward gain to track commanded arm angles with zero nominal steady-state error. The model, observer, and controllers are developed in state-space form and evaluated through linear and nonlinear simulations.

II. State Space Model

A. State and Input Definitions

The rotary arm angle is denoted by θ, and α measures the pendulum angle from its upright position. Both angles are measured, while their angular velocities complete the four-dimensional state vector. With motor voltage u as the control input, the state and output vectors are

TABLE I
Model Parameters

Quantity	Symbol	Value
Pendulum mass	m_p	0.127 kg
Pendulum length	L_p	0.33655 m
Arm mass	m_r	0.257 kg
Arm length	r_a	0.2159 m
Reflected drive inertia	J_{rot}	0.002087 kg m^2
Gear ratio	G	70
Armature resistance	R_a	$2.6\,\Omega$
Torque constant	k_t	0.00768 N m/A
Back-EMF constant	k_b	0.00768 V s/rad
Motor efficiency	η_m	0.69
Gear efficiency	η_g	0.90
Arm and pendulum damping	B_r, B_p	0.0024 N m s/rad
Gravitational acceleration	g	9.81 m/s^2

$$x = \begin{bmatrix}\theta & \alpha & \dot{\theta} & \dot{\alpha}\end{bmatrix}^T, \qquad y = \begin{bmatrix}\theta & \alpha\end{bmatrix}^T. \tag{1}$$

Near the upright equilibrium, the dynamics can be written in linear state-space form as

$$\dot{x} = Ax + Bu, \qquad y = Cx + Du. \tag{2}$$

B. Derivation of the System Matrices

Table I lists the physical and motor parameters [1], [2]. Taking the pendulum's center of mass to be at $l_p = L_p/2$, the inertias and mechanical coupling terms are

$$\begin{aligned} J_p &= \frac{m_p L_p^2}{12}, & J &= \frac{m_r r_a^2}{3} + J_{\text{rot}}, \\ M_a &= J + m_p r_a^2, & H_p &= J_p + m_p l_p^2, \\ c &= m_p l_p r_a, & h_g &= m_p l_p g. \end{aligned} \tag{3}$$

The effective motor torque coefficient and arm damping coefficient are

$$k_v = \frac{\eta_g G \eta_m k_t}{R_a}, \qquad d_v = B_r + k_v G k_b. \tag{4}$$

Linearizing about the upright equilibrium $\alpha = 0$ with $\sin\alpha \approx \alpha$ and $\cos\alpha \approx 1$ gives the coupled mechanical equations

$$\begin{bmatrix} M_a & -c \\ -c & H_p \end{bmatrix} \begin{bmatrix} \ddot{\theta} \\ \ddot{\alpha} \end{bmatrix} = \begin{bmatrix} k_v u - d_v \dot{\theta} \\ h_g \alpha - B_p \dot{\alpha} \end{bmatrix}. \tag{5}$$

Inverting the mass matrix with $\Delta = M_a H_p - c^2$ and arranging the angular accelerations in state-space form gives

$$A = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & ch_g/\Delta & -H_p d_v/\Delta & -cB_p/\Delta \\ 0 & M_a h_g/\Delta & -cd_v/\Delta & -M_a B_p/\Delta \end{bmatrix}, \quad (6)$$

$$B = \begin{bmatrix} 0 & 0 & H_p k_v/\Delta & ck_v/\Delta \end{bmatrix}^T. \quad (7)$$

Using the parameter values in Table I, the numerical system matrices are

$$A \approx \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 26.6844 & -9.4482 & -0.3055 \\ 0 & 69.4005 & -9.0917 & -0.7945 \end{bmatrix}, \quad B \approx \begin{bmatrix} 0 \\ 0 \\ 16.9843 \\ 16.3434 \end{bmatrix}. \quad (8)$$

$$C = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix}, \qquad D = 0_{2\times 1}. \quad (9)$$

C. Controllability and Observability

Controllability and observability are checked using

$$\mathcal{C} = \begin{bmatrix} B & AB & A^2B & A^3B \end{bmatrix}, \quad \mathcal{O} = \begin{bmatrix} C \\ CA \\ CA^2 \\ CA^3 \end{bmatrix}. \quad (10)$$

Here $\det \mathcal{C} \approx -1.473 \times 10^8 \neq 0$, so $\operatorname{rank} \mathcal{C} = 4$; and the rows of C and CA together form I_4, so $\operatorname{rank} \mathcal{O} = 4$. Thus, the motor input can control all four states, and the measured angles provide enough information to reconstruct the full state. The open-loop poles are

$$\operatorname{eig}(A) \approx \{0,\ 7.0915,\ -12.7738,\ -4.5604\}\ \mathrm{s}^{-1}. \quad (11)$$

The positive real pole confirms that the upright equilibrium is unstable without feedback.

III. Kalman State Estimation

A. Observer Design

The feedback controller requires both angular velocities, which are not measured directly. A Kalman observer [3] estimates them from the two measured angles and the applied motor voltage. Process and measurement noise are included in the observer design model:

$$\dot{x} = Ax + Bu + G_n w, \qquad y = Cx + v. \quad (12)$$

Here w and v are uncorrelated, zero-mean white process and measurement noises with intensity matrices Q_n and R_n. The observer is designed with

$$G_n = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 1 & 0 \\ 0 & 1 \end{bmatrix}, \qquad Q_n = 10 I_2, \qquad R_n = I_2. \quad (13)$$

Writing $\hat{x}$ for the estimated state, the innovation $\nu = y - C\hat{x}$ measures the difference between the observed angles and their model-based predictions. The observer uses this difference to correct the state estimate:

$$\dot{\hat{x}} = A\hat{x} + Bu + L\nu. \quad (14)$$

For the estimation error $\tilde{x} = x - \hat{x}$, the error dynamics are

$$\dot{\tilde{x}} = (A - LC)\tilde{x} + G_n w - Lv. \quad (15)$$

The estimation-error covariance is $P = \mathbb{E}[\tilde{x}\tilde{x}^T]$. With $W = G_n Q_n G_n^T$, its dynamics satisfy

$$\dot{P} = (A - LC)P + P(A - LC)^T + W + LR_n L^T. \quad (16)$$

Rearranging this expression by completing the square, where $L_\star = PC^T R_n^{-1}$, gives

$$\begin{aligned} \dot{P} = AP + PA^T + W - PC^T R_n^{-1} CP \\ + (L - L_\star) R_n (L - L_\star)^T. \end{aligned} \quad (17)$$

The final term is minimized by $L = L_\star$. Setting $\dot{P} = 0$ then yields the steady-state estimator Riccati equation

$$AP_e + P_e A^T - P_e C^T R_n^{-1} C P_e + W = 0, \quad (18)$$

The stabilizing solution P_e determines the Kalman gain

$$L = P_e C^T R_n^{-1} \approx \begin{bmatrix} 1.0331 & 2.6290 \\ 2.6290 & 13.6995 \\ 3.9893 & 20.2496 \\ 18.4818 & 97.2939 \end{bmatrix}. \quad (19)$$

The resulting error-dynamics poles are

$$\begin{aligned} \operatorname{eig}(A - LC) \approx \{ &-12.7726, -7.0700, \\ &-4.5614, -0.5714\}\ \mathrm{s}^{-1}. \end{aligned} \quad (20)$$

All four poles lie in the left half-plane, so the estimation error converges in the absence of noise. In the simulation, the observer receives $[u, \theta, \alpha]^T$ and is implemented with the state-space matrices

$$\begin{aligned} A_{\mathrm{obs}} &= A - LC, & B_{\mathrm{obs}} &= \begin{bmatrix} B & L \end{bmatrix}, \\ C_{\mathrm{obs}} &= I_4, & D_{\mathrm{obs}} &= 0_{4\times 3}. \end{aligned} \quad (21)$$

B. Noise Weight Comparison

The process-noise and measurement-noise weights determine how strongly the observer responds to differences between predicted and measured angles. Larger Q_n relative to R_n generally increases the influence of the measurements, while larger R_n reduces it.

To examine the effect of these weights, $R_n = I_2$ is fixed and $Q_n = qI_2$ is varied across 40 logarithmically spaced values from 0.01 to 1000. The initial error is $\tilde{x}(0) = [0.1, -0.05, 0.3, -0.2]^T$. Each noise-free error response is simulated for 60 s with a sampling interval of 0.05 s.

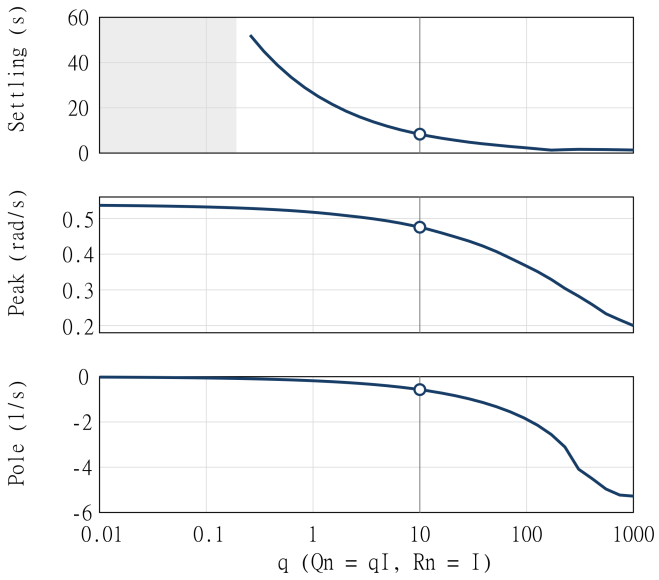


Fig. 2. LQE weight comparison. Shading marks cases with $T_s > 60$ s; circles mark the nominal setting $q = 10$.

TABLE II
KALMAN WEIGHT COMPARISON

q	T_s (s)	Peak $\lvert e_4\rvert$ (rad/s)	σ (s^{-1})
0.01	> 60	0.5367	-0.0180
10	8.35	0.4754	-0.5714
170.13	1.30	0.3292	-2.5540
1000	1.35	0.2000	-5.2790

The comparison focuses on the pendulum-speed estimation error, $e_4 = \dot{\alpha} - \widehat{\dot{\alpha}}$. Its settling time T_s is the first sampled time after which $|e_4|$ stays below 0.004 rad/s for the remainder of the test, corresponding to 2% of the initial error magnitude. The peak $|e_4|$ and the slowest error-dynamics pole, $\sigma = \max_i \mathrm{Re}\,\lambda_i(A - LC)$, are also recorded.

Figure 2 and Table II show a clear reduction in estimation error as q increases. At $q = 0.01$, the settling time exceeds 60 s and the peak error is 0.5367 rad/s. With $q = 10$, the settling time falls to 8.35 s; at $q = 1000$, it is 1.35 s and the peak error falls to 0.2000 rad/s. Although the slowest pole continues moving left, settling time increases slightly from 1.30 s at $q = 170.13$ to 1.35 s at $q = 1000$, reflecting the complete transient response relative to the settling threshold. The balancing and feedforward tracking simulations both retain $Q_n = 10I_2$ and $R_n = I_2$. Table II uses a deliberately large initial error, so the slow pole (-0.57 s^{-1} at $q = 10$) dominates its settling time. In the closed-loop tests the observer starts from $\hat{x}(0) = 0$ and the initial error is only the 5° tilt, and Fig. 3 shows the velocity errors decaying within about 2 s, so $q = 10$ was kept. A larger q would settle faster but was not re-evaluated in closed loop. Q_n and R_n are tuning weights, not identified noise statistics.

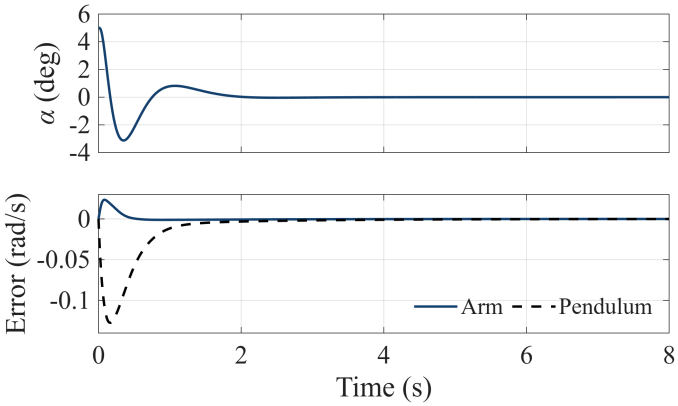


Fig. 3. Nonlinear balancing response and velocity estimation errors. Errors are true values minus estimates.

C. Velocity Estimation Results

The observer is then tested during an 8 s nonlinear balancing simulation under the LQR controller selected in Section IV, starting from $\alpha(0) = 5^\circ$ and $\hat{x}(0) = 0$. As shown in Fig. 3, the estimated angular velocities approach their true values as the pendulum recovers. During the final second, the maximum absolute errors are 3.9799×10^{-5} rad/s for the arm velocity and 1.7912×10^{-4} rad/s for the pendulum velocity.

IV. LQR STABILIZATION

A. Optimal State Feedback

Once the states can be estimated, feedback is used to return the pendulum to its upright equilibrium. The LQR controller is obtained by minimizing the infinite-horizon quadratic cost

$$J_c = \int_0^\infty (x^T Q_c x + R_c u^2)\mathrm{d}t, \qquad Q_c \succ 0, \quad R_c > 0. \tag{22}$$

The matrix Q_c penalizes state deviations, while R_c penalizes the motor voltage. Assuming a quadratic optimal value function $V(x) = x^T P_c x$, the Hamilton–Jacobi–Bellman equation becomes

$$0 = \min_u \{x^T Q_c x + R_c u^2 + 2x^T P_c (Ax + Bu)\}. \tag{23}$$

Minimizing with respect to u gives $2R_c u + 2B^T P_c x = 0$, leading to the feedback law

$$u^\star = -Kx, \qquad K = R_c^{-1} B^T P_c. \tag{24}$$

Substituting the feedback law into the Hamilton–Jacobi–Bellman equation gives the continuous-time algebraic Riccati equation

$$A^T P_c + P_c A - P_c B R_c^{-1} B^T P_c + Q_c = 0. \tag{25}$$

For the stabilizing solution P_c, the derivative of $V = x^T P_c x$ along the closed-loop trajectories is

$$\dot{V} = -x^T (Q_c + K^T R_c K) x < 0 \qquad (x \neq 0). \tag{26}$$

TABLE III
LQR WEIGHT COMPARISON

Case	q_α	R_c	Peak $\lvert\theta\rvert$ (°)	Peak $\lvert u\rvert$ (V)
1	100	1	12.57	3.76
2	300	1	12.14	4.06
3	100	0.1	11.84	10.26
4	100	10	14.09	1.91

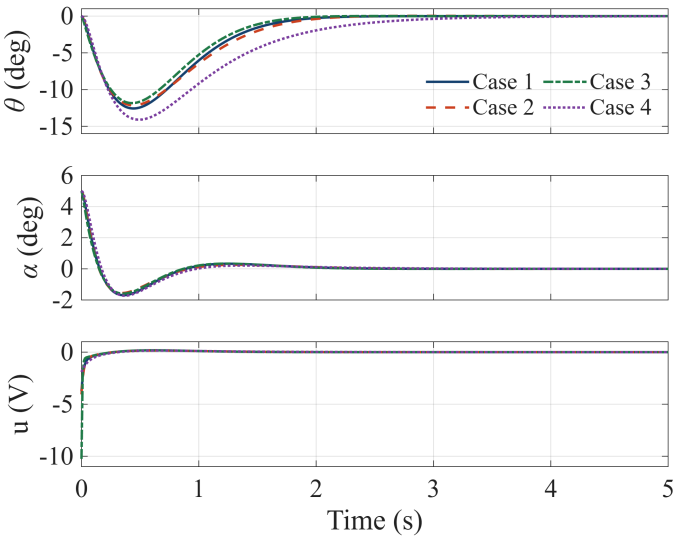


Fig. 4. LQR responses for the four weight choices in Table III, with an initial pendulum tilt of 5°. Case 2 is selected.

The negative derivative establishes asymptotic stability of the state-feedback system.

B. Weight Comparison and Selected Controller

The controller weights determine how aggressively the pendulum is balanced and how much motor voltage is used. Four choices are compared with $Q_c = \mathrm{diag}(10, q_\alpha, 1, 10)$ using the linear model. Each simulation starts from $x(0) = [0, 5\pi/180, 0, 0]^T$ and runs for 5 s. Figure 4 shows the angle and voltage responses, and Table III reports their peak values.

Increasing the pendulum-angle weight q_α from 100 to 300 reduces the peak arm rotation from 12.57° to 12.14°, while the peak voltage rises by only 0.30 V, from 3.76 to 4.06 V. Case 3 produces a slightly smaller peak arm angle of 11.84°, but requires 10.26 V. In contrast, Case 4 limits the peak voltage to 1.91 V while allowing the arm to rotate by 14.09°. Case 2 is therefore used for the remaining simulations: compared with the baseline, it reduces arm motion with a modest increase in voltage, without the much larger control demand of Case 3.

For Case 2, the feedback gain and closed-loop poles are

$$K = \begin{bmatrix} -3.1623 & 46.5279 & -3.2386 & 6.8080 \end{bmatrix}, \tag{27}$$

$$\begin{aligned} \mathrm{eig}(A - BK) \approx \{ &-56.1744, -5.4649, \\ &-2.4325 \pm j1.3163 \}\ \mathrm{s}^{-1}. \end{aligned} \tag{28}$$

In the observer-based implementation, $\hat{x}$ replaces the unavailable full state, giving $u = -K\hat{x}$. With the combined coordinates $[x^T, \tilde{x}^T]^T$, the dynamics can be written as

$$\begin{bmatrix} \dot{x} \\ \dot{\tilde{x}} \end{bmatrix} = \begin{bmatrix} A - BK & BK \\ 0 & A - LC \end{bmatrix} \begin{bmatrix} x \\ \tilde{x} \end{bmatrix}. \tag{29}$$

Because this matrix is block triangular, its eigenvalues are the poles of $A - BK$ and $A - LC$. Both sets lie in the left half-plane, so the observer and regulator form a stable LQG closed loop. The same structure explains why their gains can be designed separately.

C. Balancing Results

The nonlinear balancing simulation begins with a 5° pendulum tilt, all other physical states set to zero, and $\hat{x}(0) = 0$. Figure 3 shows the pendulum returning upright as the arm moves to recover balance. The maximum arm angle is 15.0733°, and the peak motor voltage is 1.3932 V. During the final second, from 7 to 8 s, the maximum absolute arm and pendulum angles decrease to 0.0746441° and 0.000536751°, respectively.

The voltage transient differs from the earlier linear full-state comparison because the controller initially acts on an estimated state rather than the true state. Full-state feedback gives $u(0) \approx -4.0603$ V, whereas $\hat{x}(0) = 0$ makes the initial observer-based command $u(0) = 0$.

V. REFERENCE TRACKING WITH FEEDFORWARD

A. Reference Gain Design

The arm reference r is introduced through a static feedforward gain while retaining the LQR gain K selected in Step 2. The control law is

$$u = -K\hat{x} + N_{\mathrm{bar}} r. \tag{30}$$

Thus, the tracking controller uses the same state weights $Q_c = \mathrm{diag}(10, 300, 1, 10)$ and input weight $R_c = 1$. The Kalman observer also retains $Q_n = 10I_2$ and $R_n = I_2$.

For a constant reference, write the desired steady state and input as $x_{\mathrm{ss}} = N_x r$ and $u_{\mathrm{ss}} = N_u r$. They satisfy

$$AN_x + BN_u = 0, \qquad C_\theta N_x = 1, \tag{31}$$

where $C_\theta = [1, 0, 0, 0]$ selects the arm angle. The two conditions can be solved together:

$$\begin{bmatrix} A & B \\ C_\theta & 0 \end{bmatrix} \begin{bmatrix} N_x \\ N_u \end{bmatrix} = \begin{bmatrix} 0_{4\times 1} \\ 1 \end{bmatrix}. \tag{32}$$

At steady state, the control law must supply $N_u r = -KN_x r + N_{\mathrm{bar}} r$. Therefore,

$$\begin{aligned} N_x &= \begin{bmatrix} 1 & 0 & 0 & 0 \end{bmatrix}^T, \qquad N_u = 0, \\ N_{\mathrm{bar}} &= N_u + KN_x = -3.16227766. \end{aligned} \tag{33}$$

The resulting nominal DC gain from reference to arm angle is

$$-C_\theta (A - BK)^{-1} B N_{\mathrm{bar}} = 1. \tag{34}$$

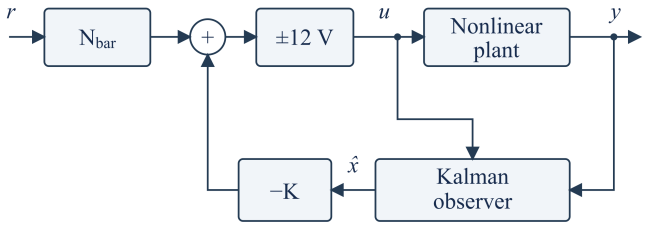


Fig. 5. Reference feedforward with LQR feedback and Kalman state estimation.

To examine convergence, define the deviation from the reference equilibrium as $\delta x = x - N_x r$. For constant r, the combined dynamics become

$$\begin{bmatrix} \dot{\delta x} \\ \dot{\tilde{x}} \end{bmatrix} = \begin{bmatrix} A - BK & BK \\ 0 & A - LC \end{bmatrix} \begin{bmatrix} \delta x \\ \tilde{x} \end{bmatrix}. \tag{35}$$

The regulator and observer poles are unchanged from the balancing design and remain in the left half-plane. Hence $\delta x \to 0$ and $\tilde{x} \to 0$, giving $\theta \to C_\theta N_x r = r$ for the nominal model.

Figure 5 shows the implementation. The reference contribution is added to the state-feedback command, and a ± 12 V limiter sets the applied voltage. This same applied voltage is supplied to the observer.

B. Reference Tracking Results

The controller is evaluated in a 40 s nonlinear Simulink simulation with $\alpha(0) = 5°$, all other physical states initially zero, and $\hat{x}(0) = 0$. The simulation is noise-free: no sensor noise or external disturbance is applied. The first 8 s allow the pendulum to recover before the arm reference changes according to

$$r(t) = \begin{cases} 0°, & 0 \le t < 8, \\ 20°, & 8 \le t < 16, \\ -15°, & 16 \le t < 24, \\ 30°, & 24 \le t < 32, \\ 0°, & 32 \le t \le 40. \end{cases} \tag{36}$$

Tracking accuracy is measured using $e = r - \theta$. For each constant-reference interval, the maximum absolute error during its final second is

$$E_{\text{tail}} = \max_{t_k \in [t_f - 1, t_f]} |e(t_k)|, \tag{37}$$

where t_f is the final sample time of that interval. Figure 6 and Table IV present the simulated response and tracking performance.

The arm reaches each commanded angle while the pendulum returns upright after each switch. The largest reference change, from $-15°$ to $30°$, produces the largest post-switch pendulum deviation, $2.7747°$, and the highest voltage peak, 2.4836 V. Across the four switched intervals, the maximum final-second arm error is $0.000796291°$, and the maximum pendulum deviation over those windows is $5.2881 \times 10^{-6}\,°$.

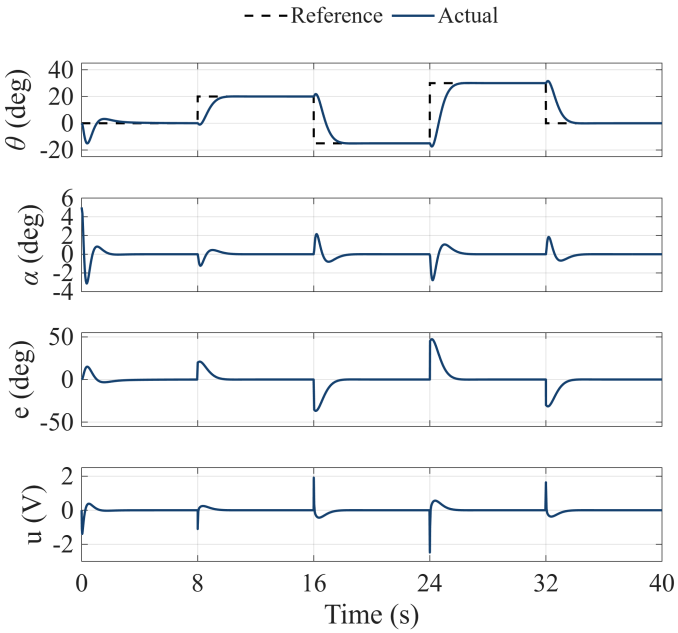


Fig. 6. Nonlinear feedforward tracking of $0°$, $20°$, $-15°$, $30°$, and $0°$.

TABLE IV
FEEDFORWARD TRACKING PERFORMANCE

Start (s)	Target (°)	Peak $\lvert\alpha\rvert$ (°)	E_{tail} (°)	Peak $\lvert u\rvert$ (V)
8	20	1.2321	7.9629×10^{-4}	1.1041
16	−15	2.1569	1.2119×10^{-4}	1.9317
24	30	2.7747	2.7664×10^{-4}	2.4836
32	0	1.8483	7.8304×10^{-5}	1.6558

The initial 0–8 s recovery interval has a final-second arm error of $0.0746441°$. Reference changes produce an immediate feedforward voltage adjustment followed by state-feedback correction. The full-run peak voltage is 2.4836 V, below the 12 V limit, so the limiter remains inactive throughout this test.

VI. Conclusion

The Kalman observer estimates the unmeasured angular velocities from the two measured angles, allowing the selected LQR controller to recover the pendulum from an initial $5°$ tilt. A reference feedforward gain extends the same feedback design to arm-angle tracking. The nonlinear simulation follows four reference changes with a maximum final-second tracking error of $0.000796291°$ and a peak motor voltage of 2.4836 V. The pendulum returns upright after each change, consistent with the closed-loop stability and nominal steady-state tracking analysis.

The design has limits. The feedforward gain removes steady-state error only for the nominal model: there is no integral action, so an unmodelled constant arm torque would

leave a steady offset, and integral augmentation was not evaluated. The observer's slowest pole ($-0.57\ \mathrm{s}^{-1}$) is slower than every regulator pole. The simulations contain no sensor noise or disturbance, and LQG has no guaranteed stability margins [4]. Future work should test noise and disturbance rejection, a faster observer and integral action before physical implementation.

REFERENCES

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APPENDIX A
STEP1.M

```
function [plant, observer] = step1
% Preserve the supplied upright linearization and four-state LQE design.
plant_parameters;
plant = struct('mp',mp,'r',r,'Lp',Lp,'lp',lp,'J',J,'Jp',Jp, ...
    'eta_g',eta_g,'G',G,'eta_m',eta_m,'kt',kt,'Ra',Ra,'kb',kb, ...
    'Bp',Bp,'Br',Br,'g',g);
Marm = J + mp*r^2;
Hpend = Jp + mp*lp^2;
couple = mp*lp*r;
hgrav = mp*lp*g;
kVoltage = eta_g*G*eta_m*kt/Ra;
dampingVoltage = Br + kVoltage*G*kb;
Delta = Marm*Hpend - couple^2;
assert(Delta > 0, 'Linearized mass matrix is singular/nonphysical.');

A = [0, 0,                      1,                       0;
     0, 0,                      0,                       1;
     0, couple*hgrav/Delta,   -Hpend*dampingVoltage/Delta, -couple*Bp/Delta;
     0, Marm*hgrav/Delta,     -couple*dampingVoltage/Delta, -Marm*Bp/Delta];
B = [0; 0; Hpend*kVoltage/Delta; couple*kVoltage/Delta];
C = [1 0 0 0; 0 1 0 0];
D = zeros(2,1);
Gnoise = [zeros(2); eye(2)];
Qn = 10*eye(2);
Rn = eye(2);
L = lqe(A,Gnoise,C,Qn,Rn);
A_obs = A-L*C;
B_obs = [B L];
C_obs = eye(4);
D_obs = zeros(4,3);
assert(rank(ctrb(A,B))==4 && rank(obsv(A,C))==4);
assert(all(real(eig(A_obs))<0));
observer = struct('A',A,'B',B,'C',C,'D',D,'Gnoise',Gnoise, ...
    'Qn',Qn,'Rn',Rn,'L',L,'A_obs',A_obs,'B_obs',B_obs,'C_obs',C_obs,'D_obs',D_obs)
    ;
fprintf('STEP1_OK | controllability 4/4 | observability 4/4\n');
disp('LQE gain L ='); disp(L);
end
```

APPENDIX B
STEP2.M

```
clc;
MA6217_intial;
load('MA6217_LQE_matrices.mat', 'A', 'B');

t = 0:0.01:5;
x0 = [0; 5*pi/180; 0; 0];

Q_values = [10 100 1 10; 10 300 1 10; 10 100 1 10; 10 100 1 10];
R_values = [1 1 0.1 10];
labels = {'Baseline', 'Higher Q_alpha', 'Lower R', 'Higher R'};
K_all = zeros(4,4);

figure(2);clf;
for i = 1:4
    Q = diag(Q_values(i,:));
    R = R_values(i);
    K = lqr(A, B, Q, R);
    K_all(i,:) = K;
    Acl = A - B*K;
    poles = eig(Acl);
    sys_cl = ss(Acl, zeros(4,1), eye(4), zeros(4,1));
    [x, tout] = initial(sys_cl, x0, t);
    u = -x*K';
    fprintf('\n%s\n', labels{i});
    fprintf('K = %s\n', mat2str(K,4));
    fprintf('Eigenvalues = %s\n', mat2str(poles.',4));
    fprintf('Max arm angle = %.2f deg\n', max(abs(x(:,1)))*180/pi);
    fprintf('Max voltage = %.2f V\n', max(abs(u)));
    subplot(3,1,1); plot(tout, x(:,1)*180/pi, 'DisplayName', labels{i}); hold on;
    subplot(3,1,2); plot(tout, x(:,2)*180/pi, 'DisplayName', labels{i}); hold on;
    subplot(3,1,3); plot(tout, u, 'DisplayName', labels{i}); hold on;
end

subplot(3,1,1); ylabel('\theta (deg)'); grid on; legend('Location','best');
subplot(3,1,2); ylabel('\alpha (deg)'); grid on;
subplot(3,1,3); ylabel('Voltage (V)'); xlabel('Time (s)'); grid on;
sgtitle('LQR Weight Comparison');

selected_case = 2;
K = K_all(selected_case,:);
disp('Selected LQR gain K:');
disp(K);

[armPartPatchHandle, armPartVertices, pendPartPatchHandle, pendPartVertices,
       drivePartPatchHandle, drivePartVertices, thetaPlotHandle, alphaPlotHandle]
       = initialiseVisualiser(l0);
```

APPENDIX C
STEP3.M

```
function design = step3(observer, baseline)
% Reference feedforward using the LQR gain selected in Step 2.
if nargin<1, [~,observer] = step1; end
if nargin<2, baseline = lqr_design(observer); end
A = observer.A;
B = observer.B;
Ctheta = observer.C(1,:);
K = baseline.K;
steady = [A B; Ctheta 0] \ [zeros(4,1); 1];
Nx = steady(1:4);
Nu = steady(5);
Nbar = Nu + K*Nx;
poles = eig(A-B*K);
dcGain = -Ctheta*((A-B*K)\(B*Nbar));
assert(all(real(poles)<0), 'Reference-tracking regulator is unstable.');
assert(abs(dcGain-1)<1e-10, 'Reference prefilter must give unit DC gain.');
design = struct('Q',baseline.Q,'R',baseline.R,'Kx',K, ...
    'Nx',Nx,'Nu',Nu,'Nbar',Nbar,'poles',poles,'dcGain',dcGain, ...
    'voltageLimit',12);
fprintf('STEP3_DESIGN_OK | K = %s | Nbar = %.8f | DC gain = %.12f\n', ...
    mat2str(K,8),Nbar,dcGain);
end
```